

LAB REPORT ON LINUX NETWORK SERVICES AND DIAGNOSTICS

Student Name	YAHAYA DANJUMA ADDRA	Registration No.	C113/26/OCO/17001
Programme/Cohort	OCO/Cohort11	Date	23/09/2026
Instructor		Score	/100
Environment	Kali Linux / Ubuntu	Submission	E-Campus

TASK 1

EVIDENCE:

```
kali@kali: ~/Documents/ICDFA_LABS/core102_lab2
Session Actions Edit View Help

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab1]
$ cd ..

(kali@kali)~[~/Documents/ICDFA_LABS]
$ mkdir -p core102_lab2

(kali@kali)~[~/Documents/ICDFA_LABS]
$ ls
accessManagement2.sh  elifstatement.shp  L3T10.png  L3T2.png  L4T3.png  LoginPage.sh  Task31L2.png  Task6.png
accessManagement.sh  existingLogin.sh  L3T11.png  L3T3.png  L4T41.png  loopcal.sh  Task3.png  Task7L2.png
Bash_script          file_manager.sh   L3T121.png L3T4.png  L4T42.png  loop.sh     Task41L2.png Task7.png
Bash_scripting       for_loop1.sh      L3T122.png L3T5.png  L4T5.png  management  Task42L2.png Task8L2.png
core102_lab1         for_loop.sh       L3T12.png  L3T6.png  L4T61.png newLogin.sh  Task43L2.png test.png
core102_lab2         forloop.sh        L3T131.png L3T72.png L4T62.png  Smart_YDee  Task4.png  while_loop.sh
createfile.sh        ICDFA_LAB1       L3T132.png L3T7.png  L4T63.png  Tas32L2.png Task51L2.png while.sh
DFIR102             if_else.sh        L3T133.png L3T82.png LAB3        Tas7NL2.png Task51NL2.png writing.sh
DFIR102_M57-jean    L3151.png        L3T134.png L3T8.png  lesson1.sh Task11.png  Task51.png
elif1.sh            L3152.png        L3T135.png L3T92.png lesson2.sh Task12L2.png Task52L2.png
elif.sh             L3153.png        L3T141.png L3T9.png  lesson3.sh Task12.png  Task52NL2.png
elif.sh.save        'L3154 (copy 1).png' L3T142.png L4T1.png  lesson4.sh Task1L2.png  Task52.png
elifstatement.sh     L3154.png        L3T1.png  L4T2.png  lesson5.sh Task2.png  Task6NL2.png

(kali@kali)~[~/Documents/ICDFA_LABS]
$ cd core102_lab2

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab2]
$ hostname
kali

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab2]
```

```
kali@kali: ~/Documents/ICDFA_LABS/core102_lab2
Session Actions Edit View Help

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab2]
$ ip -br link
lo          UNKNOWN      00:00:00:00:00:00 <LOOPBACK,UP,LOWER_UP>
eth0       UP          00:0c:29:f7:72:d2 <BROADCAST,MULTICAST,UP,LOWER_UP>

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab2]
$ ip -br addr
lo          UNKNOWN      127.0.0.1/8 ::1/128
eth0       UP          192.168.149.128/24 fe80::96d8:5426:4bfc:2254/64

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab2]
$ ip route
default via 192.168.149.2 dev eth0 proto dhcp src 192.168.149.128 metric 100
192.168.149.0/24 dev eth0 proto kernel scope link src 192.168.149.128 metric 100

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab2]
$ ip rule
0:      from all lookup local
32766:  from all lookup main
32767:  from all lookup default

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab2]
$ cat /etc/resolv.conf
# Generated by NetworkManager
search localdomain
nameserver 192.168.149.2

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab2]
$ command -v resolvectl >/dev/null && resolvectl status

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab2]
```

```
kali@kali: ~/Documents/ICDFA_LABS/core102_lab2
Session Actions Edit View Help
(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab2]
$ ip route
default via 192.168.149.2 dev eth0 proto dhcp src 192.168.149.128 metric 100
192.168.149.0/24 dev eth0 proto kernel scope link src 192.168.149.128 metric 100

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab2]
$ ip rule
0:      from all lookup local
32766:  from all lookup main
32767:  from all lookup default

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab2]
$ cat /etc/resolv.conf
# Generated by NetworkManager
search localdomain
nameserver 192.168.149.2

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab2]
$ command -v resolvectl >/dev/null && resolvectl status
(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab2]
$ command -v resolvectl >/dev/null && resolvectl status
(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab2]
$ nmcli device status 2>/dev/null || true
DEVICE  TYPE      STATE         CONNECTION
eth0     ethernet  connected    Wired connection 1
lo       loopback  connected (externally)  lo
```

RESPONSE:

Check	Server result	Client result	Interpretation
Default interface	Eth0	Connected	Ethernet interface
Private interface	Eth1	Connected	Ethernet interface
Private IPv4 address	192.168.149.128	192.168.113.129	IP address with 32-bit and 4 octets separated by dots
Default gateway	192.168.149.2	127.0.0.1	A route used in sending packets to the external network
DNS server	192.168.149.2	127.0.0.1	It helps in translating domain name to IP address

TASK 2 REPORT

EVIDENCE:

```
kali@kali: ~  
Session Actions Edit View Help  
[sudo] password for kali:  
tcpdump: verbose output suppressed, use -v[v]... for full protocol de  
listening on eth1, link-type EN10MB (Ethernet), snapshot length 26214  
^C  
0 packets captured  
0 packets received by filter  
0 packets dropped by kernel  
  
(kali@kali)-[~]  
$ sudo tcpdump -ni eth1 -c 5 arp  
[sudo] password for kali:  
tcpdump: verbose output suppressed, use -v[v]... for full protocol de  
code  
listening on eth1, link-type EN10MB (Ethernet), snapshot length 26214  
4 bytes  
04:51:15.670850 ARP, Request who-has 192.168.113.254 tell 192.168.113  
.128, length 28  
04:51:15.671134 ARP, Reply 192.168.113.254 is-at 00:50:56:e0:b9:d4, l  
ength 46  
04:51:17.110487 ARP, Request who-has 192.168.113.254 (00:50:56:e0:b9:  
d4) tell 192.168.113.1, length 46  
04:51:17.110488 ARP, Reply 192.168.113.254 is-at 00:50:56:e0:b9:d4, l  
ength 46  
05:12:49.956868 ARP, Request who-has 192.168.113.1 tell 192.168.113.1  
29, length 46  
5 packets captured  
6 packets received by filter  
0 packets dropped by kernel  
  
(kali@kali)-[~]  
$  
  
(kali@kali)-[~]  
$ ip -br link  
lo UNKNOWN 00:00:00:00:00:00 <LOOPBACK,UP,LOWER_  
UP>  
eth0 UP 00:0c:29:f7:72:d2 <BROADCAST,MULTICAS  
T,UP,LOWER_UP>  
eth1 UP 00:0c:29:f7:72:dc <BROADCAST,MULTICAS  
T,UP,LOWER_UP>  
  
(kali@kali)-[~]  
$ ip -br addr  
lo UNKNOWN 127.0.0.1/8 ::1/128  
eth0 UP 192.168.149.128/24 fe80::96d8:5426:4b  
fc:2254/64  
eth1 UP 192.168.113.128/24 fe80::e475:42ea:1d  
b0:f1f7/64  
  
(kali@kali)-[~]  
$ ping -c 1 192.168.113.128  
PING 192.168.113.128 (192.168.113.128) 56(84) bytes of data:  
64 bytes from 192.168.113.128: icmp_seq=1 ttl=64 time=1.94 ms  
  
— 192.168.113.128 ping statistics —  
1 packets transmitted, 1 received, 0% packet loss, time 0ms  
rtt min/avg/max/mdev = 1.935/1.935/1.935/0.000 ms  
  
(kali@kali)-[~]  
$
```

RESPONSE:

Packet	Source MAC	Destination MAC	Sender IP	Target IP	Meaning
ARP request	00:0c:29:f7:72:dc	ff:ff:ff:ff:ff:ff	192.168.56.1	192.168.56.100	Kali broadcasts a request asking which device owns
ARP reply	00:0c:29:8c:19:cd	00:0c:29:f7:72:dc 19	192.168.56.1	192.168.56.100	Ubuntu replies with its MAC address, allowing Kali to communicate with it.

TASK 4 REPORT

EVIDENCE:

```
kali@kali: ~/Documents/ICDFA_LABS/core102_lab2
Session Actions Edit View Help

(kali@kali)~/Documents/ICDFA_LABS/core102_lab2
$ sudo apt install -y dnsutils
[sudo] password for kali:
Note, selecting 'bind9-dnsutils' instead of 'dnsutils'
bind9-dnsutils is already the newest version (1:9.20.27-2).
The following package was automatically installed and is no longer required:
  libcrypt-dev
Use 'sudo apt autoremove' to remove it.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2250

(kali@kali)~/Documents/ICDFA_LABS/core102_lab2
$ dig example.com A

; <<>> DiG 9.20.27-2-Debian <<>> example.com A
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 34419
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 512
;; QUESTION SECTION:
;example.com.                IN      A

;; ANSWER SECTION:
example.com.                5       IN      A      104.20.23.154
example.com.                5       IN      A      172.66.147.243

;; Query time: 31 msec
;; SERVER: 192.168.149.2#53(192.168.149.2) (UDP)
;; WHEN: Tue Sep 22 11:21:18 EDT 2026
;; MSG SIZE rcvd: 72
```

```
kali@kali: ~/Documents/ICDFA_LABS/core102_lab2
Session Actions Edit View Help

(kali@kali)~/Documents/ICDFA_LABS/core102_lab2
$ dig example.com AAAA

; <<>> DiG 9.20.27-2-Debian <<>> example.com AAAA
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 3054
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 512
;; QUESTION SECTION:
;example.com.                IN      AAAA

;; ANSWER SECTION:
example.com.                5       IN      AAAA   2606:4700:10::6814:179a
example.com.                5       IN      AAAA   2606:4700:10::ac42:93f3

;; Query time: 36 msec
;; SERVER: 192.168.149.2#53(192.168.149.2) (UDP)
;; WHEN: Tue Sep 22 11:21:39 EDT 2026
;; MSG SIZE rcvd: 96
```

```
kali@kali: ~/Documents/ICDFA_LABS/core102_lab2
Session Actions Edit View Help
;; WHEN: Tue Sep 22 11:21:39 EDT 2026
;; MSG SIZE rcvd: 96

(kali@kali)~/Documents/ICDFA_LABS/core102_lab2
$ dig gmail.com MX

; <<>> DiG 9.20.27-2-Debian <<>> gmail.com MX
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 50536
;; flags: qr rd ra; QUERY: 1, ANSWER: 5, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 512
;; QUESTION SECTION:
;gmail.com. IN MX

;; ANSWER SECTION:
gmail.com. 5 IN MX 20 alt2.gmail-smtp-in.l.google.com.
gmail.com. 5 IN MX 30 alt3.gmail-smtp-in.l.google.com.
gmail.com. 5 IN MX 40 alt4.gmail-smtp-in.l.google.com.
gmail.com. 5 IN MX 5 gmail-smtp-in.l.google.com.
gmail.com. 5 IN MX 10 alt1.gmail-smtp-in.l.google.com.

;; Query time: 31 msec
;; SERVER: 192.168.149.2#53(192.168.149.2) (UDP)
;; WHEN: Tue Sep 22 11:22:05 EDT 2026
;; MSG SIZE rcvd: 161
```

```
kali@kali: ~/Documents/ICDFA_LABS/core102_lab2
Session Actions Edit View Help

;; Query time: 31 msec
;; SERVER: 192.168.149.2#53(192.168.149.2) (UDP)
;; WHEN: Tue Sep 22 11:22:05 EDT 2026
;; MSG SIZE rcvd: 161

(kali@kali)~/Documents/ICDFA_LABS/core102_lab2
$ dig -x 8.8.8.8

; <<>> DiG 9.20.27-2-Debian <<>> -x 8.8.8.8
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 43062
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 512
;; QUESTION SECTION:
;8.8.8.8.in-addr.arpa. IN PTR

;; ANSWER SECTION:
8.8.8.8.in-addr.arpa. 5 IN PTR dns.google.

;; Query time: 47 msec
;; SERVER: 192.168.149.2#53(192.168.149.2) (UDP)
;; WHEN: Tue Sep 22 11:22:24 EDT 2026
;; MSG SIZE rcvd: 73

(kali@kali)~/Documents/ICDFA_LABS/core102_lab2
```

```
kali@kali: ~/Documents/ICDFA_LABS/core102_lab2
Session Actions Edit View Help

;; Query time: 47 msec
;; SERVER: 192.168.149.2#53(192.168.149.2) (UDP)
;; WHEN: Tue Sep 22 11:22:24 EDT 2026
;; MSG SIZE rcvd: 73

(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab2]
$ host example.com
example.com has address 104.20.23.154
example.com has address 172.66.147.243
example.com has IPv6 address 2606:4700:10::6814:179a
example.com has IPv6 address 2606:4700:10::ac42:93f3
example.com mail is handled by 0 .
example.com has HTTP service bindings 1 . alpn="h2" ipv4hint=104.20.23.154,172.66.147.243 ipv6hint=2606:4700:10::6814:179a,2606:4700:10::ac42:93f3

(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab2]
$ nslookup example.com
Server:      192.168.149.2
Address:     192.168.149.2#53

Non-authoritative answer:
Name:   example.com
Address: 104.20.23.154
Name:   example.com
Address: 172.66.147.243
Name:   example.com
Address: 2606:4700:10::6814:179a
Name:   example.com
Address: 2606:4700:10::ac42:93f3
```

```
kali@kali: ~/Documents/ICDFA_LABS/core102_lab2
Session Actions Edit View Help

Address: 2606:4700:10::6814:179a
Name:   example.com
Address: 2606:4700:10::ac42:93f3

(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab2]
$ dig nonexistent-core102.example A

; <<>> DiG 9.20.27-2-Debian <<>> nonexistent-core102.example A
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NXDOMAIN, id: 7528
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 512
;; QUESTION SECTION:
;nonexistent-core102.example.  IN      A

;; AUTHORITY SECTION:
.                5      IN      SOA     a.root-servers.net. nstld.verisign-grs.com. 2026092200 1800 900 604800 86400

;; Query time: 207 msec
;; SERVER: 192.168.149.2#53(192.168.149.2) (UDP)
;; WHEN: Tue Sep 22 11:23:39 EDT 2026
;; MSG SIZE rcvd: 131

(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab2]
$ dig +trace example.com
```

```
kali@kali: ~/Documents/ICDFA_LABS/core102_lab2
Session Actions Edit View Help

(kali@kali) [~/Documents/ICDFA_LABS/core102_lab2]
$ dig +trace example.com

; <<>> DiG 9.20.27-2-Debian <<>> +trace example.com
;; global options: +cmd
.      5      IN      NS      l.root-servers.net.
.      5      IN      NS      j.root-servers.net.
.      5      IN      NS      f.root-servers.net.
.      5      IN      NS      a.root-servers.net.
.      5      IN      NS      d.root-servers.net.
.      5      IN      NS      i.root-servers.net.
.      5      IN      NS      g.root-servers.net.
.      5      IN      NS      h.root-servers.net.
.      5      IN      NS      b.root-servers.net.
.      5      IN      NS      k.root-servers.net.
.      5      IN      NS      m.root-servers.net.
.      5      IN      NS      e.root-servers.net.
.      5      IN      NS      c.root-servers.net.
.      5      IN      RRSIG   NS 8 0 518400 20261004170000 20260921160000 57780 . VxKaqUjojsf1hTgdkUAuxj38qKjdXcLisGZGYXjT
cS82aDtMDpQF+8Cl RqUpAi1Wt0BuPtDw4Qn61E78SIVtXucjOFmtCw37q/HIQ88+EBQyoHJz 8hydCgOAnd4cOEJYyBdyvXXbXD3rj24uGEb6V+FU/1nNWFO/lqRqIdhg R2GdEufW8
V2jB6ggEroLGN0a9Y6L6QyRKmd4TpLF5IabW8QF+QOHQXuc 00rCEHsB9zZoPFx/0bTkmJVUFF1RYTyhovnbB0gv5NuRoPh5/gqtuudw 7RtwizvptxbicLawPNbrZF3/Fz+cKlbn6V1
V9k1nuoBPYeFfccM9/W/J 3K5yMg==
;; Received 525 bytes from 192.168.149.2#53(192.168.149.2) in 27 ms

com.    172800  IN      NS      l.gtld-servers.net.
com.    172800  IN      NS      j.gtld-servers.net.
com.    172800  IN      NS      h.gtld-servers.net.
com.    172800  IN      NS      d.gtld-servers.net.
com.    172800  IN      NS      b.gtld-servers.net.
com.    172800  IN      NS      f.gtld-servers.net.
```

```
kali@kali: ~/Documents/ICDFA_LABS/core102_lab2
Session Actions Edit View Help

com.    172800  IN      NS      i.gtld-servers.net.
com.    172800  IN      NS      g.gtld-servers.net.
com.    172800  IN      NS      a.gtld-servers.net.
com.    172800  IN      NS      c.gtld-servers.net.
com.    172800  IN      NS      e.gtld-servers.net.
com.    86400   IN      DS      19718 13 2 8ACBB0C28F41250A80A491389424D341522D94680DA0C0291F2D3D7 71D7805A
com.    86400   IN      RRSIG   DS 8 1 86400 20261005050000 20260922040000 57780 . xw61nTX/RCAP04C/jqsTYLhfc57BSZM7yyrb-btIIZ
tIRimoNcZH/swxh p8xR4M2HFySy6Z7NGhvAd5t07WAitCq6NNWaXLY2wCNymKHYWjLmrQ1v lmVbyfqFT2mkveC6u11WTEUQMMIqi99CcF5JezadXLDIFwGzIxx8b2UB ap3s3R5onZ
vbuU00H09fZxjccnP4FFLXGvohY0tS4uh1mVnQxqhBDU5b eRzBlSFRB21cWWBC/0weS9toaxOJ9r3bBr4QT/4wZ5mh+8XGxtgo0JWX jyZS4ZC17mmv6h6igBuX6AReLRDXUpjWFG+z
riacuY5Ge3xZgcdgrtiN G1Yphg==
;; Received 1171 bytes from 198.41.0.4#53(a.root-servers.net) in 135 ms

example.com. 172800 IN NS hera.ns.cloudflare.com.
example.com. 172800 IN NS elliottns.cloudflare.com.
example.com. 86400 IN DS 2371 13 2 C988EC423E3880EB8DD8A46FE06CA230EE23F35B578D64E78B29C3E1 C83D245A
example.com. 86400 IN RRSIG DS 13 2 86400 20260926013330 20260919002330 41446 com. G91aWbKexB0EsjduEul+yJBuElxR2X/7Rrtvo
VilKe8PgF7r5tvdBfTV So0GBB7TyW5d0J9WC+uQpewItqseGw==
;; Received 506 bytes from 192.42.93.30#53(g.gtld-servers.net) in 159 ms

;; UDP setup with 2606:4700:50::adf5:3aa2#53(2606:4700:50::adf5:3aa2) for example.com failed: network unreachable.
;; no servers could be reached
;; UDP setup with 2606:4700:50::adf5:3aa2#53(2606:4700:50::adf5:3aa2) for example.com failed: network unreachable.
;; no servers could be reached
;; UDP setup with 2606:4700:50::adf5:3aa2#53(2606:4700:50::adf5:3aa2) for example.com failed: network unreachable.
;; UDP setup with 2803:f800:50::6ca2:c3e4#53(2803:f800:50::6ca2:c3e4) for example.com failed: network unreachable.
;; UDP setup with 2a06:98c1:50::ac40:20a2#53(2a06:98c1:50::ac40:20a2) for example.com failed: network unreachable.
example.com. 300 IN A 104.20.23.154
example.com. 300 IN A 172.66.147.243
example.com. 300 IN RRSIG A 13 2 300 20260923162404 20260921142404 34505 example.com. ue9ftFZJrbQSHlRYw2wwVxk8YCB89W6i
Yywa4GfbKtcy6JPkp5ckzhom LZjsPaLDb9oXZWwaf/lLD93M+Wd3rw==
;; Received 179 bytes from 162.159.44.228#53(elliott.ns.cloudflare.com) in 27 ms
```

RESPONSE:

Query	Record type	Result or status	What the result proves
example.com	A	example.com = 104.20.23.154 example.com = 172.66.147.243	The dns resolve the domain into the two addresses on the result column

Query	Record type	Result or status	What the result proves
example.com	AAAA	Ipv6 2606:4700:10::6814:179a Ipv6 2606:4700:10::ac42:93f3	The dns gives us the ipv6 for the domain “example.com”
gmail.com	MX	NOERROR 5 MX records found.	DNS successfully identified the mail servers responsible for receiving emails for gmail.com.
8.8.8.8	PTR	8.8.8.8 resolves to dns.google.	A reverse DNS lookup successfully identified the hostname associated with IP address 8.8.8.8 as dns.google.
nonexistent-core102.example	A		

TASK 4 REPORT

EVIDENCE:

```

dyahayaaddra11@ubuntu:~$ sudo tee /etc/dhcp/dhcpd.conf > /dev/null <<'EOF'
authoritative;
default-lease-time 600;
max-lease-time 7200;

subnet 192.168.56.0 netmask 255.255.255.0 {
    range 192.168.56.100 192.168.56.150;
    option routers 192.168.56.1;
    option subnet-mask 255.255.255.0;
    option broadcast-address 192.168.56.255;
    option domain-name-servers 1.1.1.1, 8.8.8.8;
}
EOF
> EOF
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
dyahayaaddra11@ubuntu:~$ sudo sed -i 's/^INTERFACESv4=.*$/INTERFACESv4="ens37"/' /etc/default/isc-dhcp-server
dyahayaaddra11@ubuntu:~$ sudo dhcpd -t -cf /etc/dhcp/dhcpd.conf
Internet Systems Consortium DHCP Server 4.4.3-P1
Copyright 2004-2022 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/
/etc/dhcp/dhcpd.conf line 12: semicolon expected.
^

```

```

Internet Systems Consortium DHCP Server 4.4.3-P1
Copyright 2004-2022 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/
Config file: /etc/dhcp/dhcpd.conf
Database file: /var/lib/dhcp/dhcpd.leases
PID file: /var/run/dhcpd.pid
dyahayaaddra11@ubuntu:~$ sudo systemctl restart isc-dhcp-server
dyahayaaddra11@ubuntu:~$ sudo systemctl --no-pager --full status isc-dhcp-server
● isc-dhcp-server.service - ISC DHCP IPv4 server
   Loaded: loaded (/usr/lib/systemd/system/isc-dhcp-server.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-09-22 20:17:06 UTC; 33s ago
  Invocation: 9ea4e99b007f44eb849f80a1f784837f
     Docs: man:dhcpd(8)
    Main PID: 409157 (dhcpd)
      Tasks: 1 (limit: 1833)
    Memory: 5.4M (peak: 6.9M)
       CPU: 73ms
    CGroup: /system.slice/isc-dhcp-server.service
            └─409157 dhcpd -user dhcpd -group dhcpd -f -4 -pf /run/dhcp-server/dhcpd.pid -cf /etc/dhcp/dhcpd.conf ens37

Sep 22 20:17:06 ubuntu dhcpd[409157]: PID file: /run/dhcp-server/dhcpd.pid
Sep 22 20:17:06 ubuntu dhcpd[409157]: Wrote 0 leases to leases file.
Sep 22 20:17:06 ubuntu sh[409157]: Wrote 0 leases to leases file.
Sep 22 20:17:06 ubuntu dhcpd[409157]: Listening on LPF/ens37/00:0c:29:8c:19:cd/192.168.56.0/24
Sep 22 20:17:06 ubuntu sh[409157]: Listening on LPF/ens37/00:0c:29:8c:19:cd/192.168.56.0/24
Sep 22 20:17:06 ubuntu sh[409157]: Sending on LPF/ens37/00:0c:29:8c:19:cd/192.168.56.0/24
Sep 22 20:17:06 ubuntu sh[409157]: Sending on Socket/fallback/fallback-net
Sep 22 20:17:06 ubuntu dhcpd[409157]: Sending on LPF/ens37/00:0c:29:8c:19:cd/192.168.56.0/24
Sep 22 20:17:06 ubuntu dhcpd[409157]: Sending on Socket/fallback/fallback-net
Sep 22 20:17:06 ubuntu dhcpd[409157]: Server starting service.

```

```

CPU: 73ms
CGroup: /system.slice/isc-dhcp-server.service
        └─409157 dhcpd -user dhcpd -group dhcpd -f -4 -pf /run/dhcp-server/dhcpd.pid -cf /etc/dhcp/dhcpd.conf ens37

Sep 22 20:17:06 ubuntu dhcpd[409157]: PID file: /run/dhcp-server/dhcpd.pid
Sep 22 20:17:06 ubuntu dhcpd[409157]: Wrote 0 Leases to leases file.
Sep 22 20:17:06 ubuntu sh[409157]: Wrote 0 leases to leases file.
Sep 22 20:17:06 ubuntu dhcpd[409157]: Listening on LPF/ens37/00:0c:29:8c:19:cd/192.168.56.0/24
Sep 22 20:17:06 ubuntu sh[409157]: Listening on LPF/ens37/00:0c:29:8c:19:cd/192.168.56.0/24
Sep 22 20:17:06 ubuntu sh[409157]: Sending on LPF/ens37/00:0c:29:8c:19:cd/192.168.56.0/24
Sep 22 20:17:06 ubuntu sh[409157]: Sending on Socket/fallback/fallback-net
Sep 22 20:17:06 ubuntu dhcpd[409157]: Sending on LPF/ens37/00:0c:29:8c:19:cd/192.168.56.0/24
Sep 22 20:17:06 ubuntu dhcpd[409157]: Sending on Socket/fallback/fallback-net
Sep 22 20:17:06 ubuntu dhcpd[409157]: Server starting service.
dyahayaaddra11@ubuntu:~$

```

```

mkdir: core102_lab2: File exists
dyahayaaddra11@ubuntu:~$ sudo tcpdump -ni ens37 -s 0 -w ~/core102_lab2/dhcp_dora.pcap 'udp port 67 or 68'
tcpdump: listening on ens37, link-type EN10MB (Ethernet), snapshot length 262144 bytes
^C13 packets captured
13 packets received by filter
0 packets dropped by kernel
dyahayaaddra11@ubuntu:~$ sudo tcpdump -nn -vv -r ~/core102_lab2/dhcp_dora.pcap 'udp port 67 or 68'
[sudo: authenticate] Password:
sudo: 'tcpdump': command not found
dyahayaaddra11@ubuntu:~$ sudo tcpdump -nn -vv -r ~/core102_lab2/dhcp_dora.pcap 'udp port 67 or 68'
reading from file /home/dyahayaaddra11/core102_lab2/dhcp_dora.pcap, 1
link-type EN10MB (Ethernet), snapshot length 262144
20:31:06.935624 IP (tos 0x0, ttl 128, id 8909, offset 0, flags [none]
, proto UDP (17), length 333)
    192.168.113.1.68 > 192.168.113.254.67: [udp sum ok] BOOTP/DHCP, R
equest from 00:50:56:c0:00:01, length 305, xid 0xa1d4f2e4, Flags [non
e] (0x0000)
        Client-IP 192.168.113.1
        Client-Ethernet-Address 00:50:56:c0:00:01
        Vendor-rfc1048 Extensions
            Magic Cookie 0x63825363
            DHCP-Message (53), length 1: Request
            Client-ID (61), length 7: ether 00:50:56:c0:00:01
            Hostname (12), length 9: "SmartYDee"
            FQDN (81), length 12: "SmartYDee"
            Vendor-Class (60), length 8: "MSFT 5.0"
            Parameter-Request (55), length 15:
                Subnet-Mask (1), Default-Gateway (3), Domain-Name-Serve
r (6), Domain-Name (15)

```

```

Client-Ethernet-Address 00:0c:29:f7:72:dc
Vendor-rfc1048 Extensions
  Magic Cookie 0x63825363
  DHCP-Message (53), length 1: Offer
  Server-ID (54), length 4: 192.168.113.129
  Lease-Time (51), length 4: 600
  Subnet-Mask (1), length 4: 255.255.255.0
  BR (28), length 4: 192.168.56.255
  Default-Gateway (3), length 4: 192.168.56.1
  Domain-Name-Server (6), length 8: 1.1.1.1,8.8.8.8
20:45:20.703174 IP (tos 0x10, ttl 128, id 0, offset 0, flags [none],
proto UDP (17), length 328)
  0.0.0.0.68 > 255.255.255.255.67: [udp sum ok] BOOTP/DHCP, Request
  from 00:0c:29:f7:72:dc, length 300, xid 0x1e7b283a, Flags [none] (0x
0000)
  Client-Ethernet-Address 00:0c:29:f7:72:dc
  Vendor-rfc1048 Extensions
    Magic Cookie 0x63825363
    DHCP-Message (53), length 1: Request
    Server-ID (54), length 4: 192.168.113.129
    Requested-IP (50), length 4: 192.168.56.100
    Hostname (12), length 4: "kali"
    Parameter-Request (55), length 13:
      Subnet-Mask (1), BR (28), Time-Zone (2), Default-Gatewa
y (3)
      Domain-Name (15), Domain-Name-Server (6), Unknown (119)
, Hostname (12)
      Netbios-Name-Server (44), Netbios-Scope (47), MTU (26),
Classless-Static-Route (121)
      NTP (42)
20:45:20.703174 IP (tos 0x10, ttl 16, id 0, offset 0, flags [none], p

```

```

Vendor-rfc1048 Extensions
  Magic Cookie 0x63825363
  DHCP-Message (53), length 1: Request
  Client-ID (61), length 7: ether 00:50:56:c0:00:01
  Hostname (12), length 9: "SmartYDee"
  FQDN (81), length 12: "SmartYDee"
  Vendor-Class (60), length 8: "MSFT 5.0"
  Parameter-Request (55), length 15:
    Subnet-Mask (1), Default-Gateway (3), Domain-Name-Serve
r (6), Domain-Name (15)
    Router-Discovery (31), Static-Route (33), Vendor-Option
(43), Netbios-Name-Server (44)
    Netbios-Node (46), Netbios-Scope (47), Unknown (119), C
lassless-Static-Route (121)
    Classless-Static-Route-Microsoft (249), Unknown (252),
URL (114)
20:46:07.268238 IP (tos 0x10, ttl 16, id 0, offset 0, flags [none], p
roto UDP (17), length 328)
  192.168.113.254.67 > 192.168.113.1.68: [udp sum ok] BOOTP/DHCP, R
eply, length 300, xid 0x16038a0e, Flags [none] (0x0000)
  Client-IP 192.168.113.1
  Your-IP 192.168.113.1
  Server-IP 192.168.113.254
  Client-Ethernet-Address 00:50:56:c0:00:01
  Vendor-rfc1048 Extensions
    Magic Cookie 0x63825363
    DHCP-Message (53), length 1: ACK
    Server-ID (54), length 4: 192.168.113.254
    Lease-Time (51), length 4: 1800
    Subnet-Mask (1), length 4: 255.255.255.0
(i-search)''

```

RESPONSE:

I make Ubuntu the DHCP server.

I give Ubuntu a fixed address: 192.168.56.1.

I tell Ubuntu: "When a computer joins this network, give it an available IP address between 192.168.56.100 and 192.168.56.150."

I check that the DHCP configuration has no mistakes.

I start the DHCP service.

So, the simple idea is:

Kali: “I need an IP address.”

Ubuntu: “Okay, take 192.168.56.100.”

TASK 5 REPORT

EVIDENCE:

```
et:1 http://kali.download/kali kali-rolling/main amd64 isc-dhcp-clie
t amd64 4.4.3-P1-8 [1,099 kB]
et:2 http://kali.download/kali kali-rolling/main amd64 isc-dhcp-comm
n amd64 4.4.3-P1-8 [118 kB]
etched 1,217 kB in 4s (333 kB/s)
electing previously unselected package isc-dhcp-client.
Reading database ... 425125 files and directories currently installe
.)
preparing to unpack .../isc-dhcp-client_4.4.3-P1-8_amd64.deb ...
unpacking isc-dhcp-client (4.4.3-P1-8) ...
electing previously unselected package isc-dhcp-common.
preparing to unpack .../isc-dhcp-common_4.4.3-P1-8_amd64.deb ...
unpacking isc-dhcp-common (4.4.3-P1-8) ...
etting up isc-dhcp-client (4.4.3-P1-8) ...
etting up isc-dhcp-common (4.4.3-P1-8) ...
rocessing triggers for man-db (2.13.1-1) ...
rocessing triggers for kali-menu (2025.4.3) ...

(kali@kali)-[~]
$ dhclient --version
isc-dhclient-4.4.3-P1

(kali@kali)-[~]
$ sudo dhclient -r eth1

(kali@kali)-[~]
$ sudo dhclient -v eth1
Internet Systems Consortium DHCP Client 4.4.3-P1
copyright 2004-2022 Internet Systems Consortium.
all rights reserved.
```

RESPONSE:

DORA stage	Source address	Destination address	UDP ports	Key evidence
Discover	0.0.0.0:68	255.255.255.255	68-67	DHCP-Message: Discover; Client MAC 00:0c:29:f7:72:dc
Offer	192.168.113.129:67	192.168.56.100	67-68	Offer; offered IP 192.168.56.100; lease 600 seconds
Request	0.0.0.0:68	255.255.255.255	68-67	Request; Requested-IP: 192.168.56.100

DORA stage	Source address	Destination address	UDP ports	Key evidence
Acknowledge	192.168.113.129:67	192.168.56.100	67-68	DHCP-Message: ACK; IP 192.168.56.100; gateway 192.168.56.1; DNS 1.1.1.1, 8.8.8.8

TASK 6 REPORT

EVIDENCE:

RESPONSE:

Symptom	First command	Confirming evidence	Safe corrective action
No IPv4 address on private adapter	Ip -br addr show	Has no	Bring the interface up and renew
Address exists but server cannot be pinged	Eth1	192.168.56.x/24 address or the interface is down	sudo ip link set eth1 up the sudo dhclient -v eth1
IP connectivity works but names fail	Ip route[	The client has an IP address, but the route to 192.168.56.0/24 is missing or incorrect.	Check or erify the private interface and route; if appropriate, add the missing connected route with sudo ip route add 192.168.56.0/24 dev eth1
DNS server responds but one name is NXDOMAIN	dig example.com	ping 1.1.1.1 works but DNS lookup fails, indicating a DNS/resolver problem.	Verify the dns configuration and ensure the dns server is working
DHCP client sends Discover but receives no Offer	sudo tcpdump -ni eth1 'udp port 67 or 68'	Repeated DHCPDISCOVER packets are visible, but no DHCPOFFER is received.	Verify that isc-dhcp server is running and listening on ens7 and both the client and host sever are connected to the sane private network
A neighbour entry shows FAILED	Ip neigh show	The target ip shows FAILED indicating ARP resolution issue	Verify the interface/link and subnet, then flush the stale neighbour entry with sudo ip neigh flush dev eth1 and retry.

TASK 7 REPORT

EVIDENCE:

```
(kali㉿kali)-[~/Documents/ICDFA_LABS]
$ cd core102_lab2

(kali㉿kali)-[~/Documents/ICDFA_LABS/core102_lab2]
$ ls

(kali㉿kali)-[~/Documents/ICDFA_LABS/core102_lab2]
(kali㉿kali)-[~/Documents/ICDFA_LABS/core102_lab2]
$ nano snapshot.sh

(kali㉿kali)-[~/Documents/ICDFA_LABS/core102_lab2]
$ chmod +x snapshot.sh

(kali㉿kali)-[~/Documents/ICDFA_LABS/core102_lab2]
$ bash -n snapshot.sh

(kali㉿kali)-[~/Documents/ICDFA_LABS/core102_lab2]
$ ./snapshot.sh
Saved: network_snapshot_20260923_055011.txt

(kali㉿kali)-[~/Documents/ICDFA_LABS/core102_lab2]
$ nano snapshot.sh

(kali㉿kali)-[~/Documents/ICDFA_LABS/core102_lab2]
$ ls -l snapshot_*.txt
ls: cannot access 'snapshot_*.txt': No such file or directory

(kali㉿kali)-[~/Documents/ICDFA_LABS/core102_lab2]
$ ls -l network_snapshot_*.txt
-rw-rw-r-- 1 kali kali 1480 Sep 23 05:50 network_snapshot_20260923_055011.txt
```

```

$ ls -l network_snapshot_*.txt
-rw-rw-r-- 1 kali kali 1480 Sep 23 05:50 network_snapshot_20260923_05
5011.txt

(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab2]
$ sed -n '1,120p' network_snapshot_*.txt
CORE102 NETWORK SNAPSHOT
Generated: 2026-09-23T05:50:11-04:00
Host: kali
User: kali

== INTERFACES ==
lo                UNKNOWN          00:00:00:00:00:00 <LOOPBACK,UP,LOWER_
UP>
eth0              UP              00:0c:29:f7:72:d2 <BROADCAST,MULTICAS
T,UP,LOWER_UP>
eth1              UP              00:0c:29:f7:72:dc <BROADCAST,MULTICAS
T,UP,LOWER_UP>
lo                UNKNOWN          127.0.0.1/8 ::1/128
eth0              UP              192.168.149.128/24 fe80::96d8:5426:4b
fc:2254/64
eth1              UP              192.168.113.128/24 192.168.56.100/24
fe80::e475:42ea:1db0:f1f7/64

== ROUTES ==
default via 192.168.149.2 dev eth0 proto dhcp src 192.168.149.128 met
ric 102
192.168.56.0/24 dev eth1 proto kernel scope link src 192.168.56.100
192.168.113.0/24 dev eth1 proto kernel scope link src 192.168.113.128
metric 103

```

```

== ROUTES ==
default via 192.168.149.2 dev eth0 proto dhcp src 192.168.149.128 metric 102
192.168.56.0/24 dev eth1 proto kernel scope link src 192.168.56.100
192.168.113.0/24 dev eth1 proto kernel scope link src 192.168.113.128 metric 103
192.168.149.0/24 dev eth0 proto kernel scope link src 192.168.149.128 metric 102

== NEIGHBOURS ==
192.168.113.254 dev eth1 lladdr 00:50:56:ee:10:f7 STALE
192.168.113.129 dev eth1 lladdr 00:0c:29:8c:19:cd STALE
192.168.149.130 dev eth0 lladdr 00:0c:29:8c:19:c3 STALE
192.168.56.1 dev eth1 lladdr 00:0c:29:8c:19:cd STALE
192.168.149.254 dev eth0 lladdr 00:50:56:ea:ff:4d STALE
192.168.149.2 dev eth0 lladdr 00:50:56:fc:63:12 REACHABLE

== RESOLVER ==
nameserver 1.1.1.1
nameserver 8.8.8.8

== LISTENING SOCKETS ==
Netid State Recv-Q Send-Q Local Address:Port Peer Address:Port
udp UNCONN 0 0 0.0.0.0:68 0.0.0.0:*

(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab2]
$

```

RESPONSE:

Section	Command	Purpose
Interfaces	<code>ip -br link / ip -br addr</code>	Shows network interfaces, status and IP addresses
Routes	<code>ip route</code>	Shows routing table
Neighbours	<code>ip neigh show</code>	Shows ARP/neighbor cache
Resolver	<code>cat /etc/resolv.conf</code>	Shows DNS resolver configuration
Listening sockets	<code>ss -tuln</code>	Shows TCP/UDP listening ports

TASK 8 REPORT

EVIDENCE:

```

> ^C
dyahayaaddra11@ubuntu:~$ sudo systemctl stop isc-dhcp-server
[sudo: authenticate] Password:
dyahayaaddra11@ubuntu:~$ sudo systemctl status isc-dhcp-server
--no-pager
o isc-dhcp-server.service - ISC DHCP IPv4 server
   Loaded: loaded (/usr/lib/systemd/system/isc-dhcp-server.service; enabled; preset: enabled)
   Active: inactive (dead) since Wed 2026-09-23 10:55:10 UTC; 44s ago
     Duration: 14h 38min 4.435s
  Invocation: 9ea4e99b007f44eb849f80a1f784837f
     Docs: man:dhcpd(8)
    Process: 409157 ExecStart=/bin/sh -ec CONFIG_FILE=/etc/dhcp/dhcpd.conf; if [ -f /etc/ltsp/dhcpd.conf ]; then CONFIG_FILE=/etc/ltsp/dhcpd.conf; fi; [ -e /var/lib/dhcp/dhcpd.leases ] || touch /var/lib/dhcp/dhcpd.leases; chown root:dhcpd /var/lib/dhcp /var/lib/dhcp/dhcpd.leases; chmod 775 /var/lib/dhcp ; chmod 664 /var/lib/dhcp/dhcpd.leases; exec dhcpd -user dhcpd -group dhcpd -f -4 -pf /run/dhcp-server/dhcpd.pid -cf $CONFIG_FILE $INTERFACESv4 (code=killed, signal=TERM)
   Main PID: 409157 (code=killed, signal=TERM)
  Mem peak: 6.9M (swap: 3.6M)
    CPU: 813ms

Sep 23 10:46:12 ubuntu dhcpd[409157]: DHCPREQUEST for 192.16...37
Sep 23 10:46:12 ubuntu dhcpd[409157]: DHCPACK on 192.168.56....37
Sep 23 10:50:13 ubuntu dhcpd[409157]: DHCPREQUEST for 192.16...37
Sep 23 10:50:13 ubuntu dhcpd[409157]: DHCPACK on 192.168.56....37
Sep 23 10:52:35 ubuntu dhcpd[409157]: DHCPREQUEST for 192.16...9.

```

```

192.168.56.1/24 fe80::20c:29ff:fe8c:19cd/64
dyahayaaddra11@ubuntu:~$ sudo ss -ltnp | grep ':67 ' || echo "N
o DHCP listener"
No DHCP listener
dyahayaaddra11@ubuntu:~$ sudo ip addr del 192.168.56.1/24 dev "
$LAN_IFACE"
Cannot find device ""
dyahayaaddra11@ubuntu:~$ sudo ip addr del 192.168.56.1/24 dev e
ns37
dyahayaaddra11@ubuntu:~$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNK
NOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfif
o_fast state UP group default qlen 1000
    link/ether 00:0c:29:8c:19:c3 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c298c19c3
    inet 192.168.149.130/24 metric 100 brd 192.168.149.255 scop
e global dynamic ens33
        valid_lft 1552sec preferred_lft 1552sec
    inet6 fe80::20c:29ff:fe8c:19c3/64 scope link proto kernel_l
l
        valid_lft forever preferred_lft forever
3: ens37: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfif
o_fast state UP group default qlen 1000
    link/ether 00:0c:29:8c:19:cd brd ff:ff:ff:ff:ff:ff

```

RESPONSE:

In this LAB I learned how to stop my isc-dhcp server after installing the dhcp server and ran the necessary application it was installed to do. dhcp server enables auto-ip assignment to devices on the same network. I finally deleted the ip address I added to my private network purposely for the LAB.